



E550 FW Solution Evaluation VO.1

Henry

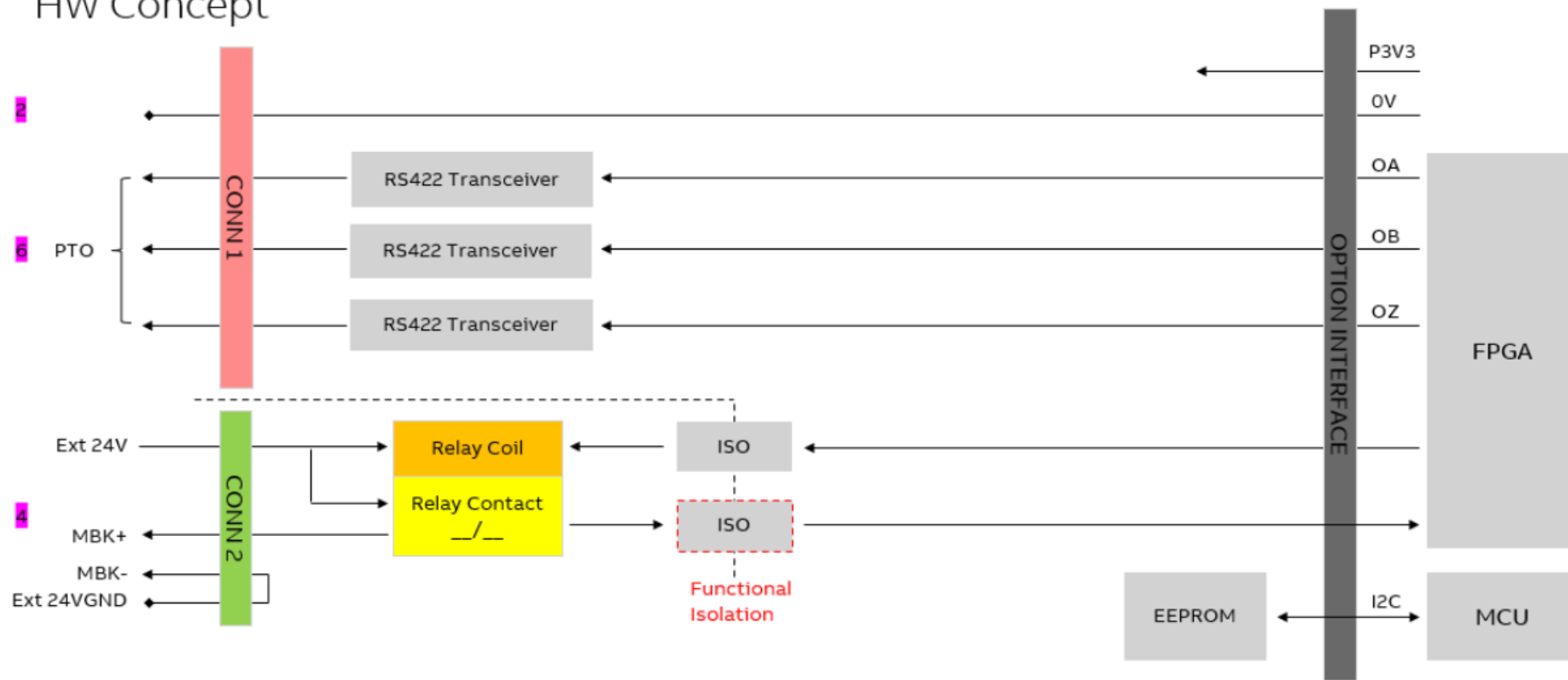
ENGINEERED
TO OUTFIT

Extension Module 1

1. HW concept determination

Extension Module 1

HW Concept





Extension Module 1

FW feature Concept

MCU:

- 1. I2C interface to EEPROM
- 2. PTO user IF : need to be clarified, reuse PTI parameter? COE object? PN?

4	5	DEC-U16	Pulse Output Source Selection
4	6	DEC-U16	Pulse Output Phase
4	7	DEC-U16	Z Pulse Output Polarity
4	8	DEC-U16	Number of Output Pulses per Revolution
4	9	DEC-U16	Electronic Gear Ratio Numerator of Pulse Output
4	10	DEC-U16	Electronic Gear Ratio Denominator of Pulse Output

- 3. BK Selection user IF : need to be Clarify



Extension Module 1

FW feature Concept

FPGA:

1. PTO module:

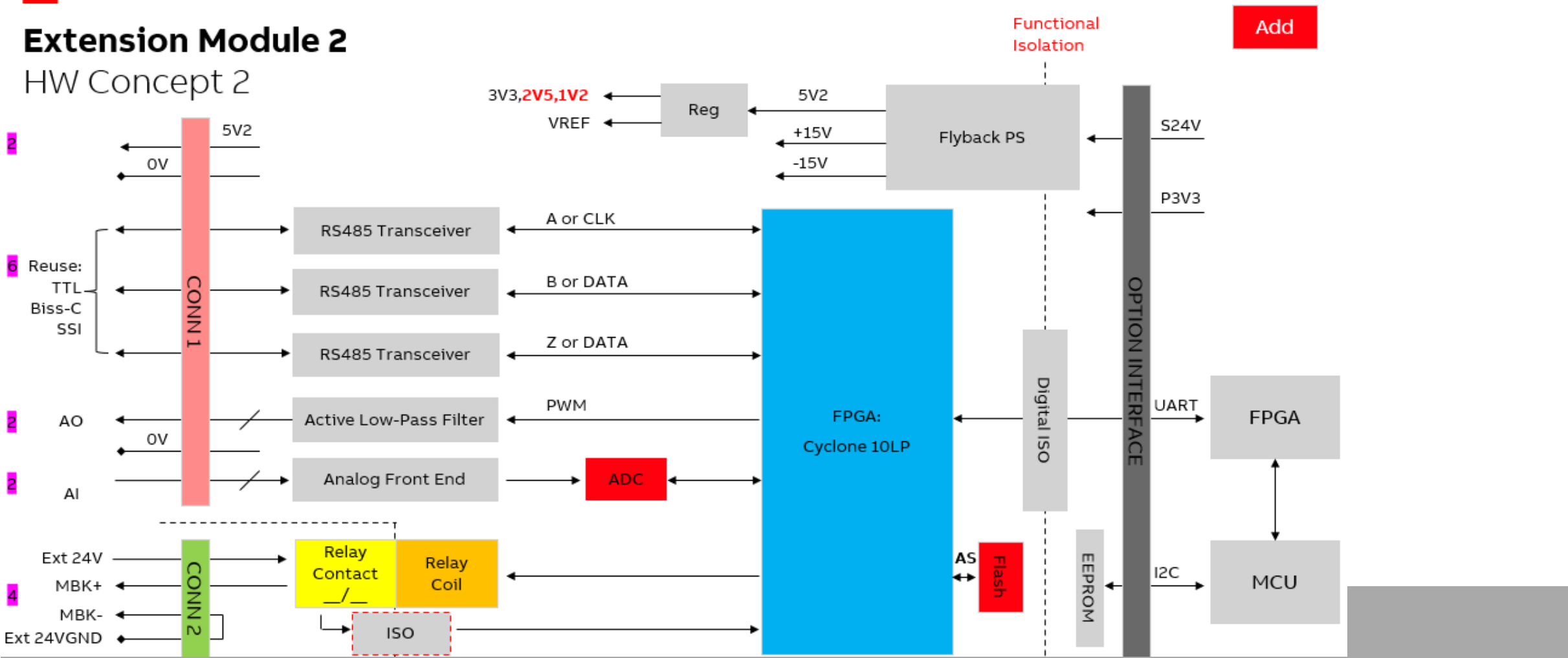
▼	abc u PTO:PTO	728 (487)
---	------------------	-----------

2. BK Signal handler

Extension Module 2

Extension Module 2

HW Concept 2



Extension Module 2

FW feature Concept

MCU: (Same with Module 1 part)

1. I2C interface to EEPROM
2. PTO user IF : need to be clarified, reuse PTI parameter? Align COE object, PN?

4	5	DEC-U16	Pulse Output Source Selection
4	6	DEC-U16	Pulse Output Phase
4	7	DEC-U16	Z Pulse Output Polarity
4	8	DEC-U16	Number of Output Pulses per Revolution
4	9	DEC-U16	Electronic Gear Ratio Numerator of Pulse Output
4	10	DEC-U16	Electronic Gear Ratio Denominator of Pulse Output

3. BK Selection user IF : need to be clarified



Extension Module 2

FW feature Concept

MCU: (Module 2 specific part)

1. AI/AO user IF : need to be clarified

Analog input use cases	Analog output use cases
- Used as speed/torque setpoint for local speed/torque control. - Used as speed/torque limit source. - Used as speed/torque forward source. - Used as remote analog input for controller to save controller cost. - Used as one input source when programming function is ready. - Used as other unidentified use cases.	- Used as outputting different signal (e.g. speed, torque, DC bus voltage, etc.) status to controller or 3rd party devices. - Used as remote analog output for controller to save controller cost. - Used as one output source when programming function is ready. - Used as other unidentified use cases.

Based on current information, we propose to have the technical evaluation or reserve with below AI/AO specifications. So that we can easily add the required functions with less hardware change and mainly firmware upgrade when we clarified the requirements from customers.

What feature is necessary for Analog input?

Item1: DI SPD function need to be added like PTI? P5.02 parameter name?

New Project_2026-04-02_15_45_46*

1. E530-PT 220V_1

E530-PT 220V(1.6.0.8)

192.168.0.1

Online

Add to ▾

Copy to ▾

📄

📄

📄

📄

📄

📄

📄

📄

Search by ID or NAME

🔍

☐ ID	NAME	PREPARED VALUE	ACTIVE
☐ P.05.00	AltSpeedTransformationCoefficient	-	-
☐ P.05.01	JogSpeed	-	-
☐ P.05.02	InternalSpeedCommand1	-	-
☐ P.05.03	InternalSpeedCommand2	-	-
☐ P.05.04	InternalSpeedCommand3	-	-
☐ P.05.05	InternalSpeedCommand4	-	-
☐ P.05.06	InternalSpeedCommand5	-	-
☐ P.05.07	InternalSpeedCommand6	-	-
☐ P.05.08	InternalSpeedCommand7	-	-
☐ P.05.09	AccelerationTimeOfSpeedCommand	-	-
☐ P.05.10	DecelerationTimeOfSpeedCommand	-	-
☐ P.05.11	SCurveSetting	-	-
☐ P.05.12	TorqueFeedforwardSourceSelection	-	▾
☐ P.05.13	ZeroClampSpeedThreshold	-	▾

PARAMETER GROUP

Customized(0+0)/100

Customized(0+0)/100

Customized3(0+0)/100

Auto tuning parameters

Prepared Value # Empty

Active Value # Default Value

Prepared Value # Active Value

Prepared Value # Default Value

All Parameters

dPO Monitoring parameters

P01 Basic parameters

P02 Gain parameters

P03 Tuning parameters

P04 Position control parameters

P05 Speed control parameters

P06 Torque control parameters

P07 Extension control parameters

ID	NAME	PREPARED VALUE	ACTIVE VALUE				AX
P 05.01	JogSpeed	-	100	rpm	100	0	6000
P 05.02	InternalSpeedCommand	-	100	rpm	100	-6000	6000
P 05.09	AccelerationTimeOfSpeedCommand	-	100	ms/1000rpm	100	0	65535
P 05.10	DecelerationTimeOfSpeedCommand	-	100	ms/1000rpm	100	0	65535
P 05.11	SCurveSetting	-	0	ms	0	0	1000
P 05.12	TorqueFeedforwardSourceSelection	-	3		3	0	3
P 05.13	ZeroClampSpeedThreshold	-	20	rpm	20	0	500
P 05.14	ZSPDSignalOutputCheckWindow	-	20	rpm	20	0	500
P 05.15	SARRSignalOutputCheckWindow	-	1000	rpm	1000	0	6000
P 05.16	SCMPSignalOutputCheckWindow	-	10	rpm	10	10	500
P 05.17	MaximumSpeedLimit	-	6000	rpm	6000	0	10000

- 0: Disable;
- 1: SON (Servo On);
- 2: FRST (Fault Reset);
- 3: POT (Positive Limit);
- 4: NOT (Negative Limit);
- 5: EMGS (Emergency Stop);
- 6: JOGP (Jog In Positive Direction);
- 7: JOGN (Jog In Negative Direction);
- 8: JOGEN(Jog Enable);
- 9: INHP (Pulse Command Inhibit);
- 10: CLR (Position Error Clear);
- 11: GAIN (Gain Switch);
- 12: GEAR1 (Electronic Gear Selection 1);
- 13: GEAR2 (Electronic Gear Selection 2);
- 14: SPD1 (Speed Command Source Selection 1);
- 15: SPD2 (Speed Command Source Selection 2);
- 16: SPD3 (Speed Command Source Selection 3);
- 17: SDIR (Speed Command Direction Selection In Speed Mode) ;
- 18: ZCLMP (Zero Speed Clamp);
- 19: TCMD (Torque Command Source Selection);
- 20: TDIR (Torque Command Direction Selection In Torque Mode);
- 21: MSEL1 (Control Mode Selection 1);
- 22: MSEL2 (Control Mode Selection 2);
- 23: SPL1 (Speed Limit Source Selection 1);
- 24: SPL2 (Speed Limit Source Selection 2);
- 25: TQL1 (Torque Limit Source Selection 1);
- 26: TQL2 (Torque Limit Source Selection 2);
- 27: SHOM (Homing Start);
- 28: ORG (Origin Signal);
- 29: POS1 (FP Command Source Selection 1);
- 30: POS2 (FP Command Source Selection 2);
- 31: POS3 (FP Command Source Selection 3);
- 32: POS4 (FP Command Source Selection 4);
- 33: CRG (Start Positioning In FP Control Mode)

- 0: Disable;
- 1: SON (Servo On);
- 2: FRST (Fault Reset);
- 3: POT (Positive Limit);
- 4: NOT (Negative Limit);
- 5: EMGS (Emergency Stop);
- 6: JOGP (Jog In Positive Direction);
- 7: JOGN (Jog In Negative Direction);
- 8: JOGEN(Jog Enable);
- 9: FSTOP(Force Stop);
- 11: GAIN (Gain Switch);
- 23: SPL1 (Speed Limit Source Selection 1);
- 24: SPL2 (Speed Limit Source Selection 2);
- 25: TQL1 (Torque Limit Source Selection 1);
- 26: TQL2 (Torque Limit Source Selection 2);
- 28: ORG (Origin Signal);
- 34: TP1 (Touch Probe 1);
- 35: TP2 (Touch Probe 2);

Extension Module 2

FW feature Concept

MCU: (Module 2 specific part)

1. AI/AO user IF : need to be clarified

Analog input use cases	Analog output use cases
1. Used as speed/torque setpoint for local speed/torque control. 2. Used as speed/torque limit source. 3. Used as speed/torque forward source. 4. Used as remote analog input for controller to save controller cost. 5. Used as one input source when programming function is ready. 6. Used as other unidentified use cases.	1. Used as outputting different signal (e.g. sp etc.)status to controller or 3rd party devices 2. Used as remote analog output for controller, to save controller cost. 3. Used as one output source when programming function is ready. 4. Used as other unidentified use cases.

Based on current information, we propose to have the technical evaluation or reserve with below AI/AO specifications. So that we can easily add the required functions with less hardware change and mainly firmware upgrade when we clarified the requirements from customers.

What feature is necessary for Analog input?

Item2: add 0—1 selection for EP driver

7.8.2 转矩速度限制

转矩速度限制命令源:

- 外部模拟量 AI1
- 内部速度限制命令, 共三组

在转矩控制模式下, 通过DI信号选择速度限制命令源。需将两个DI 分别配置为 SPL1 和 SPL2 信号, 并选择 DI有效时的电平逻辑。

信号类型	信号名称	功能码	触发方式	描述
DI	SPL1	23	电平	选择速度限制命令源。
DI	SPL2	24	电平	

速度限制命令源与 DI 信号的对应关系如下:

速度限制命令来源	SPL2	SPL1
内部速度限制命令 1	0	0
外部模拟量 AI1	0	1
内部速度限制命令 2	1	0
内部速度限制命令 3	1	1

注: 0= 无效; 1= 有效

7.9.2 一般转矩限制

一般转矩限制命令源:

- 外部模拟量 AI2
- 内部转矩限制命令, 共三组

通过DI信号选择转矩限制命令来源。需将两个DI 分别配置为 TQL1 和 TQL2 信号, 并选择DI 有效时的电平逻辑。

信号类型	信号名称	功能码	触发方式	描述
DI	TQL1	25	电平	选择转矩限制命令源。
DI	TQL2	26	电平	

转矩限制命令源与 DI 信号的对应关系如下:

转矩限制命令	TQL2	TQL1
内部转矩限制命令 1	0	0
外部模拟量 AI2	0	1
内部转矩限制命令 2	1	0
内部转矩限制命令 3	1	1

注: 0= 无效; 1= 有效。

速度限制命令源与 DI 信号的对应关系如下:

速度限制命令	SPL2	SPL1
内部速度限制命令 1	0	0
内部速度限制命令 2	1	0
内部速度限制命令 3	1	1

注: 0= 无效; 1= 有效

转矩限制命令源与 DI 信号的对应关系如下:

转矩限制命令	TQL2	TQL1
内部转矩限制命令 1	0	0
内部转矩限制命令 2	1	0
内部转矩限制命令 3	1	1

注: 0= 无效; 1= 有效

Extension Module 2

FW feature Concept

MCU: (Module 2 specific part)

1. AI/AO user IF : need to be clarified

Analog input use cases	Analog output use cases
1. Used as speed/torque setpoint for local speed/torque control. 2. Used as speed/torque limit source. 3. Used as speed/torque forward source. 4. Used as remote analog input for controller to save controller cost. 5. Used as one input source when programming function is ready. 6. Used as other unidentified use cases.	1. Used as outputting different signal (e.g. speed, torque, DC bus voltage, etc.)status to controller or 3rd party devices. 2. Used as remote analog output for controller to save controller cost. 3. Used as one output source when programming function is ready. 4. Used as other unidentified use cases.

Based on current information, we propose to have the technical evaluation or reserve with below AI/AO specifications. So that we can easily add the required functions with less hardware change and mainly firmware upgrade when we clarified the requirements from customers.

What feature is necessary for Analog input?

Item3: How shall we define the parameter descriptions?

5	12	DEC-U16	Torque Feedforward Source Selection	RW	NVOL	0	3	3	1		Para	immediately	YES	YES	YES	EC	List	0: None; 1: Internal; 2: Reserved; 3: Object 0x60B2
5	12	DEC-U16	Torque Feedforward Source Selection	RW	NVOL	0	3	3	1		Para	immediately	YES	YES	YES	EC_400	List	0: None; 1: Internal; 2: Reserved; 3: Object 0x60B2



Extension Module 2

FW feature Concept

MCU: (Module 2 specific part)

1. AI/AO user IF : need to be clarified

Analog input use cases	Analog output use cases
1. Used as speed/torque setpoint for local speed/torque control. 2. Used as speed/torque limit source. 3. Used as speed/torque forward source. 4. Used as remote analog input for controller to save controller cost. 5. Used as one input source when programming function is ready. 6. Used as other unidentified use cases.	1. Used as outputting different signal (e.g. speed, torque, DC bus voltage, etc.)status to controller or 3rd party devices. 2. Used as remote analog output for controller to save controller cost. 3. Used as one output source when programming function is ready. 4. Used as other unidentified use cases.

Based on current information, we propose to have the technical evaluation or reserve with below AI/AO specifications. So that we can easily add the required functions with less hardware change and mainly firmware upgrade when we clarified the requirements from customers.

What feature is necessary for Analog input?

Item4-5:

Analog input for controller/programming function, the function need to be clarified?





Extension Module 2

FW feature Concept

MCU: (Module 2 specific part)

1. AI/AO user IF : need to be clarified

Analog input use cases	Analog output use cases
1. Used as speed/torque setpoint for local speed/torque control. 2. Used as speed/torque limit source. 3. Used as speed/torque forward source. 4. Used as remote analog input for controller to save controller cost. 5. Used as one input source when programming function is ready. 6. Used as other unidentified use cases.	1. Used as outputting different signal (e.g. speed, torque, DC bus voltage, etc.)status to controller or 3rd party devices. 2. Used as remote analog output for controller to save controller cost. 3. Used as one output source when programming function is ready. 4. Used as other unidentified use cases.

Based on current information, we propose to have the technical evaluation or reserve with below AI/AO specifications. So that we can easily add the required functions with less hardware change and mainly firmware upgrade when we clarified the requirements from customers.

What feature is necessary for Analog output?

Item1:

The objects need to be clarified, it may affect the interface between drive and extension module?

Item2-3: The function need to be clarified.





Extension Module 2

FW feature Concept

MCU: (Module 2 specific part)

1. AI/AO user IF : need to be clarified

Analog input use cases	Analog output use cases
1. Used as speed/torque setpoint for local speed/torque control. 2. Used as speed/torque limit source. 3. Used as speed/torque forward source. 4. Used as remote analog input for controller to save controller cost. 5. Used as one input source when programming function is ready. 6. Used as other unidentified use cases.	1. Used as outputting different signal (e.g. speed, torque, DC bus voltage, etc.)status to controller or 3rd party devices. 2. Used as remote analog output for controller to save controller cost. 3. Used as one output source when programming function is ready. 4. Used as other unidentified use cases.

Based on current information, we propose to have the technical evaluation or reserve with below AI/AO specifications. So that we can easily add the required functions with less hardware change and mainly firmware upgrade when we clarified the requirements from customers.

Does the AI/AO function require production line process support?



Extension Module 2

FW feature Concept

MCU: (Module 2 specific part)

1. Full closed loop:

Step 1: a pre-study/bench-mark is necessary from my side. One of our competitor need to be chosen.

Step 2: Solution implementation

Extension Module 2

FW feature Concept

FPGA: (Module 2 specific part)

1. The internal interface between extension module and drive (FPGA to FPGA) : **need system design input, Global solution?**
 - Interface definition
 - Communication protocol

FW request:

3rd-part motor control related time sensitive information must update every 62.5us to full fill E530 control Arc.

Full-close loop related information depends on outcome of pre-study/Bench-mark. But will not alter the basic control structure of E530.

Extension Module 2

FW feature Concept

Risk: (Module 2 specific part)

1. FPGA size

Current size(E550 G3A)

Quartus Prime Version	23.1std.0 Build 991 11/28/2023 SC Lite Edition
Revision Name	PmsmCtrlTop_Mars
Top-level Entity Name	PmsmCtrlTop_Mars
Family	Cyclone 10 LP
Device	10CL025YU256I7G
Timing Models	Final
Total logic elements	17 671 / 24 624 (72 %)
Total registers	13423
Total pins	94 / 151 (62 %)
Total virtual pins	0
Total memory bits	132,488 / 608,256 (22 %)
Embedded Multiplier 9-bit elements	102 / 132 (77 %)
Total PLLs	1 / 4 (25 %)

Total Size													
24624													
Current Size	Endat		PTO	interface			Full closed loop				BK	SUM	Percent
	Endat protocol	Endat data buffer		Protocol	Frame handler	Others???	Pos Regulator	config IF	Protection/Faults	Unit handler			
17671	3000	600	728	0	0	0	600	60	100	100	10	22869	92.87%

Extension Module 2

FW feature Concept

Risk: (Module 2 specific part)

2. 3rd-part motor/full closed loop/Al feedforward performance.

Based on the overall solution design, the performance will theoretically be lower than Inovance.

3. BK performance.

BK performance will be different. Module 1-Module 2.

Summary

FW feature

1. **System design for the whole solution is needed as an input for FW arc/function design.**
2. **Some of the functions and user IF need to be clarified for further design and efforts evaluation.**
3. **Production line requirement input?**
4. **From a product design perspective, how should we deal with the existing risk points?**

ABB